Autoencoders and representation learning

Fashion-MNIST

LECTURE 11 LECTURE NOTES

Applications of Machine Learning

BSc Mathematics of Artificial Intelligence · Third year

Where we are

Lecture	What it built	What it needed
7	a first network, and what a layer computes	a label for every image
8	the training loop, written out in five lines	a label for every image
9	a deep network that trains, by initialisation and normalisation	a label for every image
10	a network that stops overfitting: early stopping, weight decay, dropout, augmentation	a label for every image
11	**a network that learns from the images alone**	**no labels**

OUTSIDE THE BOOK

This lecture is taught from its notes rather than from Géron, and is examined on the same terms as every other. The notes are the primary source; the deck is the lecture.

Today

1. The task: a catalogue of photographs and no labels — find the similar ones, group them, flag the odd ones
2. **The mathematics** — a linear autoencoder with squared loss recovers the principal subspace of Lecture 6
3. **The method** — undercomplete autoencoders in PyTorch; what the loss sees; the code as a representation, measured on retrieval and clustering; denoising; anomaly detection
4. The variational autoencoder, for context, and where representations go from here
5. **Design exercise** — a fashion marketplace's "see similar" feature, in nine steps, with the pitfalls and how to address them

Open Lecture 11 in Google Colab

The setting: a catalogue nobody labelled

A marketplace holds millions of product photographs uploaded by sellers. The category field is free text, often empty, sometimes wrong. Three things the product team wants from the photographs alone:

- **Similar items** — given one garment, the ten most like it, in milliseconds, across the whole catalogue
- **Groups** — a first structure for a catalogue that has none
- **The odd ones** — uploads that are not what the catalogue holds, flagged before a shopper sees them

✓ Every method so far needed a label per row. None of these questions comes with one. The network of today learns from the inputs by trying to *reproduce* them.

The data, and the one rule about labels

Fashion-MNIST	Count
Images to fit on	12,000
Validation images	5,000
Test images, the catalogue	10,000
Pixels per image	784
Classes	10

The same split as Lectures 7 and 8, so the numbers compare. Twelve thousand images are few enough to train every network here on a CPU.

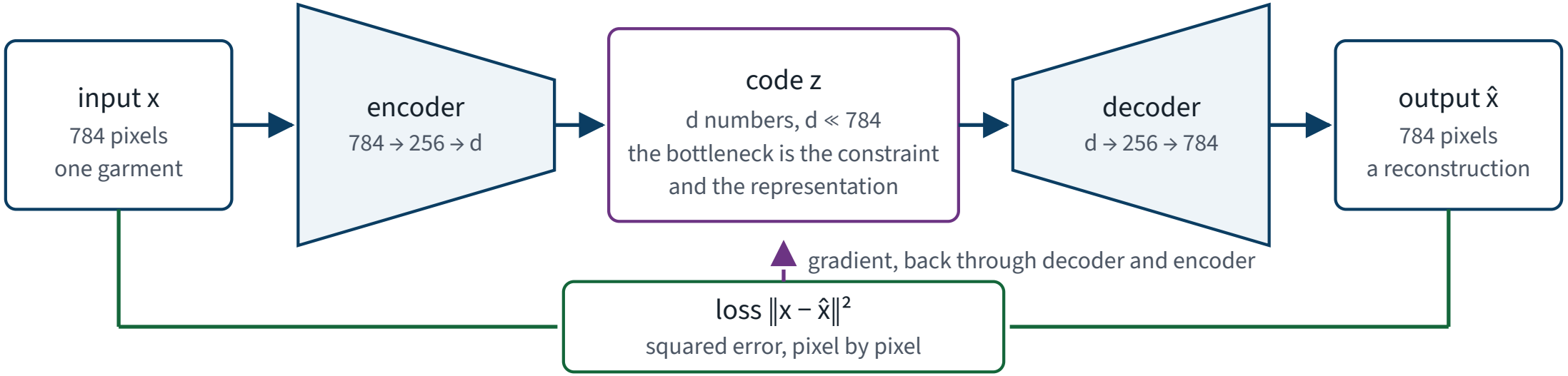
THE LABELS ARE LOADED, AND USED FOR EXACTLY ONE THING

Scoring. Nothing is fitted on them. Precision at ten, the clusters' purity and the anomaly detector's AUROC all read the labels after the fact, to say whether a representation learned without them means anything.

An autoencoder: compress, then reconstruct

Encoder, code, decoder

the target is the input; no label appears anywhere



After training, the encoder alone is kept: image in, d numbers out.

The target is the input. The network is trained to return what it was given, through a code too small to hold it — so it has to learn what the images have in common.

The objective

$$\mathbf{z} = f_{\text{enc}}(\mathbf{x}) \in \mathbb{R}^d, \qquad \hat{\mathbf{x}} = f_{\text{dec}}(\mathbf{z}) \in \mathbb{R}^n$$

$$L = \frac{1}{m} \sum_{i=1}^{m} \left\| \mathbf{x}^{(i)} - \hat{\mathbf{x}}^{(i)} \right\|_2^2$$

- $n = 784$ pixels in, n pixels out; d numbers in between — the **code**
- Squared loss on pixels: the same loss as least squares, with the input standing in for the label
- Trained by the five-line loop of Lecture 8; nothing in the loop knows there is no label

The whole design question is what stops the network from learning the identity.

Why a bottleneck: a constraint in the sense of Lecture 10

Lecture 10's regularisers — weight decay, dropout, augmentation — all work the same way: they *shrink the set of functions the network can reach*, so that fitting the training data no longer means memorising it. An autoencoder needs the same thing, for a sharper reason: with $d = n$ the identity is reachable, the loss is zero, and nothing was learned.

Constraint	What it forbids	Name
$d \ll n$	**copying: 784 numbers cannot pass through 32**	**undercomplete**
corrupt the input, keep the target clean	copying: what comes in is not what must go out	denoising
penalise the code	using every coordinate for every image	sparse

✓ Today: the first two. The first is the one the mathematics is about.

The question for the mathematics

Make the encoder and the decoder *linear*, keep the squared loss, and train by gradient descent. What does the network learn?

> **THE CLAIM**
>
> Exactly what Lecture 6 computed with the SVD: the best such network reconstructs through the projection onto the span of the leading d right singular vectors, and the error it leaves is the tail of the spectrum, $\sum_{j>d} \sigma_j^2$.

The result is worth having twice over. It says what an autoencoder is, in the one case where the answer is closed-form; and it says what the non-linear one is *for* — the part of the error that no projection can remove.

Step 1 · The model, with the shapes tracked

One photograph per row, as in Lecture 6; the data centred on its mean.

EXAMINABLE

$$\mathbf{Z} = \tilde{\mathbf{X}}\mathbf{A}, \quad \hat{\mathbf{X}} = \mathbf{Z}\mathbf{B} = \tilde{\mathbf{X}}\mathbf{A}\mathbf{B}, \quad E(\mathbf{A}, \mathbf{B}) = \left\|\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\mathbf{A}\mathbf{B}\right\|$$

Object	Shape	Role
$\tilde{\mathbf{X}}$	$m \times n$	the centred images, one per row
$\mathbf{A}$	$n \times d$	the encoder: n pixels to d numbers
$\mathbf{Z}$	$m \times d$	the codes, one per row
$\mathbf{B}$	$d \times n$	the decoder: d numbers to n pixels

No orthogonality, no constraint on $\mathbf{A}$ or $\mathbf{B}$ at all. Gradient descent sees $2nd$ free numbers.

Step 2 · Every reconstruction lies in a d-dimensional subspace

Row i of $\hat{\mathbf{X}}$ is $\mathbf{z}^{(i)}\mathbf{B}$ — a combination of the d rows of $\mathbf{B}$, with the d entries of the code as coefficients.

$$\hat{\mathbf{x}}^{(i)} \in S := \text{rowspace}(\mathbf{B}) \subseteq \mathbb{R}^n, \qquad \dim S \leq d$$

Whatever the encoder does, the decoder can only put a reconstruction *somewhere in* S. The encoder chooses where; the decoder fixes the subspace. This is the only place the bottleneck enters the argument, and it enters as a dimension count.

Step 3 · The closest point of a subspace is the projection

Let $\mathbf{P}_S$ be the orthogonal projector onto S. For any $\mathbf{y} \in S$, split the error into the part orthogonal to S and the part inside it:

$$\tilde{\mathbf{x}} - \mathbf{y} = \underbrace{(\tilde{\mathbf{x}} - \mathbf{P}_S\tilde{\mathbf{x}})}_{\perp S} + \underbrace{(\mathbf{P}_S\tilde{\mathbf{x}} - \mathbf{y})}_{\in S}$$

$$\|\tilde{\mathbf{x}} - \mathbf{y}\|^2 = \|\tilde{\mathbf{x}} - \mathbf{P}_S\tilde{\mathbf{x}}\|^2 + \|\mathbf{P}_S\tilde{\mathbf{x}} - \mathbf{y}\|^2$$

Pythagoras, because the two pieces are orthogonal. The second term is never negative, so

$$\|\tilde{\mathbf{x}} - \mathbf{y}\|^2 \;\geq\; \|\tilde{\mathbf{x}} - \mathbf{P}_S\tilde{\mathbf{x}}\|^2 \qquad \text{for every } \mathbf{y} \in S.$$

Step 4 · Sum over the rows: the loss is bounded by a projection's

Apply step 3 to every row, with $\mathbf{y} = \hat{\mathbf{x}}^{(i)}$:

$$E(\mathbf{A}, \mathbf{B}) = \sum_i \|\tilde{\mathbf{x}}^{(i)} - \hat{\mathbf{x}}^{(i)}\|^2 \;\geq\; \sum_i \|\tilde{\mathbf{x}}^{(i)} - \mathbf{P}_S \tilde{\mathbf{x}}^{(i)}\|^2$$

$$= \left\|\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\mathbf{W}\mathbf{W}^\intercal\right\|_F^2$$

where $\mathbf{W}$ is an orthonormal basis of S, so that $\mathbf{P}_S = \mathbf{W}\mathbf{W}^\intercal$. If $\dim S < d$, pad $\mathbf{W}$ to d columns: a larger subspace can only lower the error.

✓ The right-hand side is *exactly* Lecture 6's objective $E(\mathbf{W})$, with $\mathbf{W}^\intercal\mathbf{W} = \mathbf{I}_d$.

Step 5 · Recall what Lecture 6 proved about that objective

PCA VIA THE SVD, RESTATED

With $\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and $\sigma_1 \geq \sigma_2 \geq \ldots$, over all $\mathbf{W}$ with orthonormal columns,

$$\min_{\mathbf{W}} \left\| \tilde{\mathbf{X}} - \tilde{\mathbf{X}}\mathbf{W}\mathbf{W}^\top \right\|_F^2 = \sum_{j>d} \sigma_j^2, \qquad \text{attained at } \mathbf{W} = \mathbf{V}_d,$$

the first d columns of $\mathbf{V}$.

Seven lines there: Pythagoras turns the error into $\|\tilde{\mathbf{X}}\|_F^2 - \operatorname{tr}(\mathbf{W}^\top \mathbf{C}\mathbf{W})$, the SVD diagonalises $\mathbf{C} = \tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$, and d units of weight spent on the largest σ_j^2 maximise the trace. Nothing in it is needed again today except the result.

Step 6 · The bound is attained by a linear autoencoder

Steps 4 and 5 together: for every $\mathbf{A}$ and $\mathbf{B}$, $E(\mathbf{A}, \mathbf{B}) \geq \sum_{j>d} \sigma_j^2$. Now choose

$$\mathbf{A} = \mathbf{V}_d, \quad \mathbf{B} = \mathbf{V}_d^\mathsf{T} \quad \Longrightarrow \quad \mathbf{AB} = \mathbf{V}_d \mathbf{V}_d^\mathsf{T}$$

$$E(\mathbf{V}_d, \mathbf{V}_d^\mathsf{T}) = \sum_{j>d} \sigma_j^2$$

The encoder computes the principal components; the decoder puts them back. The lower bound is reached, so it is the minimum: $\min_{\mathbf{A},\mathbf{B}} E(\mathbf{A}, \mathbf{B}) = \sum_{j>d} \sigma_j^2$.

Step 7 · The subspace is unique; the code is not

At any optimum the reconstruction is $\tilde{\mathbf{X}}\mathbf{V}_d\mathbf{V}_d^\top$ — step 3 is an equality only when $\hat{\mathbf{x}}^{(i)} = \mathbf{P}_S\tilde{\mathbf{x}}^{(i)}$, and step 5 is an equality only on the principal subspace when the σ_j^2 around position d are distinct. But for any invertible $d \times d$ matrix $\mathbf{T}$,

$$(\mathbf{A}\mathbf{T})\,(\mathbf{T}^{-1}\mathbf{B}) = \mathbf{A}\mathbf{B}$$

gives the same reconstruction with a different code, $\mathbf{Z}\mathbf{T}$. Gradient descent lands on *one* of these: the principal subspace, in some basis, not the principal components themselves.

✓ That is the claim to check numerically: the error, and the subspace — not the coordinates.

The derivation, in seven lines

Reproduce this from the statement of the problem, with Lecture 6's result quoted at line 5.

EXAMINABLE

1. $E(\mathbf{A}, \mathbf{B}) = \|\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\mathbf{A}\mathbf{B}\|_F^2$, $\mathbf{A}$ is $n \times d$, $\mathbf{B}$ is $d \times n$
   the linear autoencoder, unconstrained

2. $\hat{\mathbf{x}}^{(i)} = \mathbf{z}^{(i)}\mathbf{B} \in S = \text{rowspace}(\mathbf{B})$, $\dim S \leq d$
   every reconstruction lies in one subspace

3. $\|\tilde{\mathbf{x}} - \mathbf{y}\|^2 \geq \|\tilde{\mathbf{x}} - \mathbf{P}_S\tilde{\mathbf{x}}\|^2$ for $\mathbf{y} \in S$
   Pythagoras: the projection is the closest point

4. $E(\mathbf{A}, \mathbf{B}) \geq \|\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\mathbf{W}\mathbf{W}^\intercal\|_F^2$, $\mathbf{W}^\intercal\mathbf{W} = \mathbf{I}_d$
   sum over rows; $\mathbf{W}$ an orthonormal basis of S

5. $\min_{\mathbf{W}} \|\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\mathbf{W}\mathbf{W}^\intercal\|_F^2 = \sum_{j>d} \sigma_j^2$ at $\mathbf{W} = \mathbf{V}_d$
   Lecture 6: PCA via the SVD

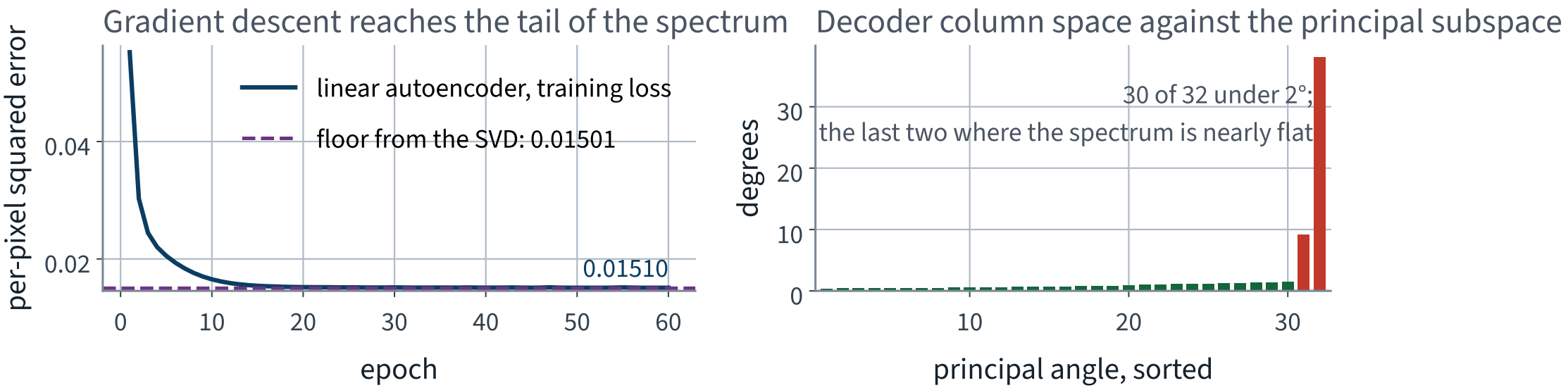
6. $\mathbf{A} = \mathbf{V}_d, \mathbf{B} = \mathbf{V}_d^\intercal$ gives $E = \sum_{j>d} \sigma_j^2$
   the bound is attained, so it is the minimum

7. $\min_{\mathbf{A},\mathbf{B}} E = \sum_{j>d} \sigma_j^2$, reconstruction through $\mathbf{V}_d\mathbf{V}_d^\intercal$; code unique up to $\mathbf{T}$
   PCA, reached by gradient descent

What the result buys you

- **A floor you can compute before training.** $\sum_{j>d} \sigma_j^2$ comes from one SVD. Any linear autoencoder that reports less has a bug; one that reports much more has not converged.
- **A definition of what the non-linear model is for.** Whatever it gains over the floor is the part of the error that no projection can remove — curvature in the data that a subspace cannot follow.
- **The bottleneck is the whole constraint.** At $d = n$ the tail is empty, the floor is zero, the identity is reached, and the code is a rotation of the pixels. Nothing learned, no error raised.
- **The code is a representation only up to a change of basis.** Distances in it are not the distances between principal components; they are those distances under $\mathbf{T}$. Retrieval in a learned code is retrieval in a metric nobody chose.

Checked numerically: gradient descent reaches the floor



Floor from the SVD 0.01501; the linear autoencoder after 60 epochs 0.01510. Thirty of the 32 principal angles are under two degrees. The last two are where σ_{32}^2 and σ_{33}^2 differ by about one percent: the loss barely prefers one direction to the other, so neither does the optimiser — step 6, seen from inside.

The same subspace, three ways of saying so

Quantity, at $d = 32$	Linear autoencoder	Rank-32 projection
Training error, per pixel	0.01510	0.01501 (the floor)
Test error, per pixel	0.01523	0.01519
Principal angles under 2°	30 of 32	—
σ_{32}^2 and σ_{33}^2, per pixel	0.000230 and 0.000227 — about one percent apart	
R^2 of the code regressed on the PCA scores	**0.9979 ✓**	—

The last row is step 7: the code is not the principal components, but a linear map of them explains it almost entirely. Same subspace, different basis.

The projection is computed in closed form; the network by 60 epochs of Adam on 12,000 images. The derivation says they must agree, and they do.

The undercomplete autoencoder, in PyTorch

```python
class AutoEncoder(nn.Module):
    def __init__(self, d, hidden=256):
        super().__init__()
        self.enc = nn.Sequential(nn.Linear(784, hidden), nn.ReLU(),
                                 nn.Linear(hidden, d))
        self.dec = nn.Sequential(nn.Linear(d, hidden), nn.ReLU(),
                                 nn.Linear(hidden, 784), nn.Sigmoid())

    def forward(self, x):
        return self.dec(self.enc(x))
```

- $784 \to 256 \to d \to 256 \to 784$; a sigmoid on the output, because a reconstruction is an image in $[0, 1]$
- `enc` and `dec` are attributes on purpose: the code is read by calling the encoder alone
- At $d = 32$: 419,120 parameters — the two 784-wide layers are almost all of them

The loop is the one you already own

```
1  for epoch in range(EPOCHS):
2      net.train()
3      for xb in batches(X_fit):           # no labels in the loader
4          opt.zero_grad()                 # 1
5          out  = net(xb)                  # 2
6          loss = F.mse_loss(out, xb)      # 3  the target is the input
7          loss.backward()                 # 4
8          opt.step()                      # 5
```

One change from Lecture 8: line 3 compares the output with the *input*. Everything about training deep networks — initialisation, the optimiser, the schedule, the regularisers — carries over unchanged, because nothing in it ever depended on the target being a label.

Twenty epochs of Adam at 10^{-3}, batches of 128, 12,000 images: a CPU is enough.

Run it at $d = 32$

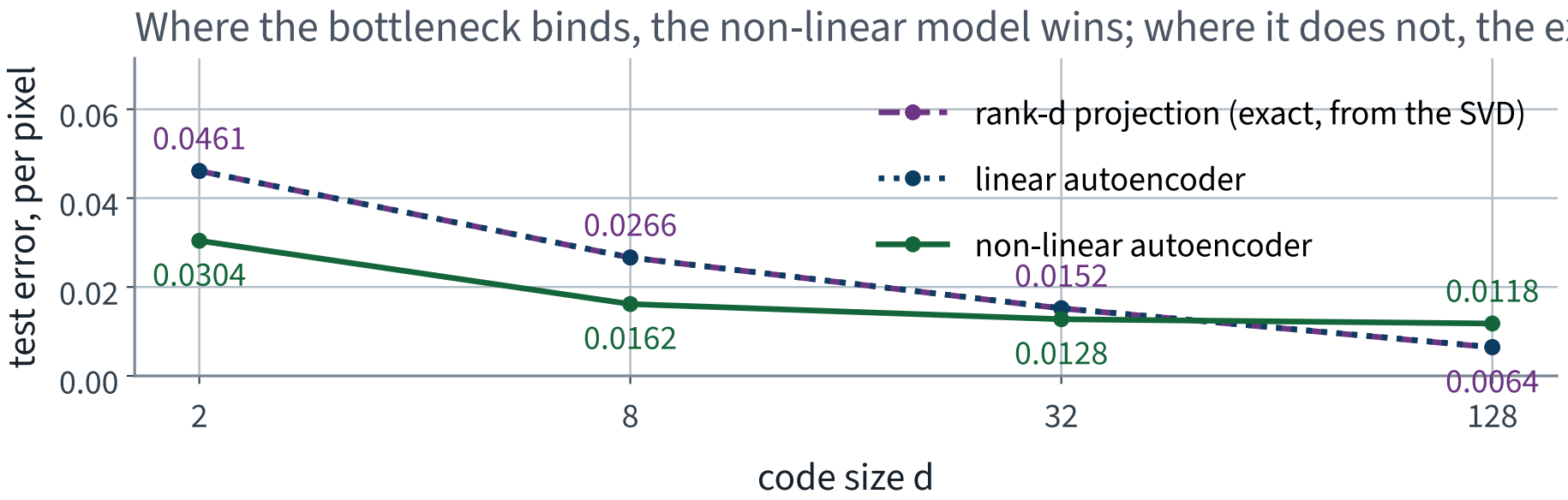
Model, test error per pixel	Error
Rank-32 projection — the best any *linear* model can do	0.01519
Linear autoencoder	0.01523
Non-linear autoencoder, 419,120 parameters	**0.01275 ✓**

About a sixth below the floor.

That sixth is the quantity the derivation defined: the error a subspace cannot remove, which a curved map can. It is the entire justification for the two ReLU layers.

Training error 0.01188, validation 0.01276: a small gap, the kind Lecture 10 was about, and small because the bottleneck is doing the regulariser's job.

The code size, swept



The linear network tracks the exact projection at every d. The non-linear one wins where the bottleneck binds — and at $d = 128$ it *loses*: 0.01179 against 0.00641.

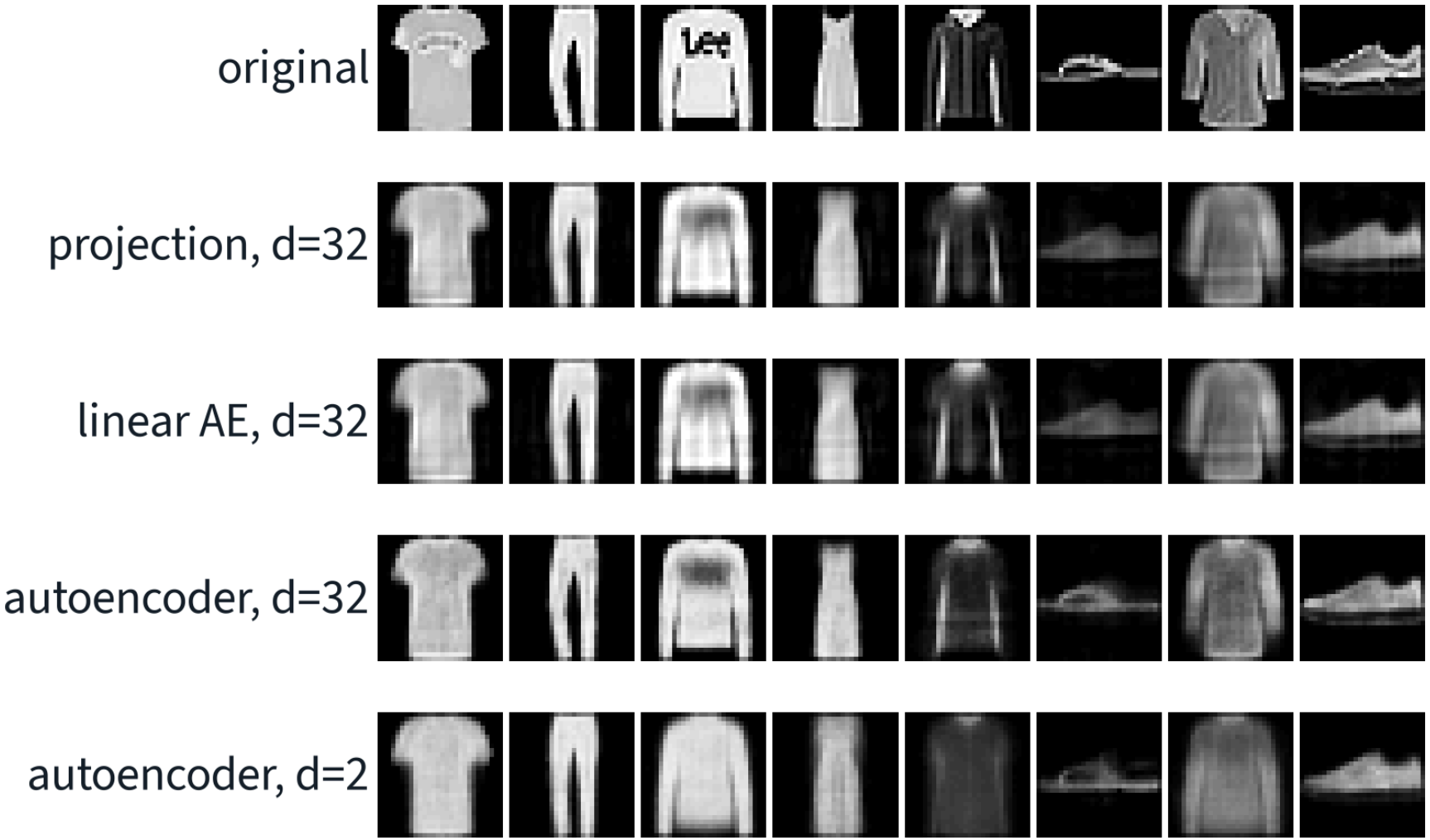
Reading the last column first

d	Projection (exact)	Linear autoencoder	Non-linear autoencoder
2	0.04610	0.04611	**0.03041 ✓**
8	0.02662	0.02665	**0.01617 ✓**
32	0.01519	0.01523	**0.01275 ✓**
128	**0.00641 ✓**	0.00649	**✗ 0.01179**

FAILURE CONDITION — A TRAINED APPROXIMATION AGAINST AN EXACT OPTIMUM

The projection is the exact minimum of its class, from one SVD. The network is a *trained* approximation to the minimum of a larger class, and at $d = 128$ its budget — one hidden layer of 256, twenty epochs — leaves it above what the SVD gets in closed form. More capacity is not more accuracy until the optimiser has found it.

What the loss sees



Rows two and three are the derivation in pictures: the same reconstruction. Row five is what two numbers buy.

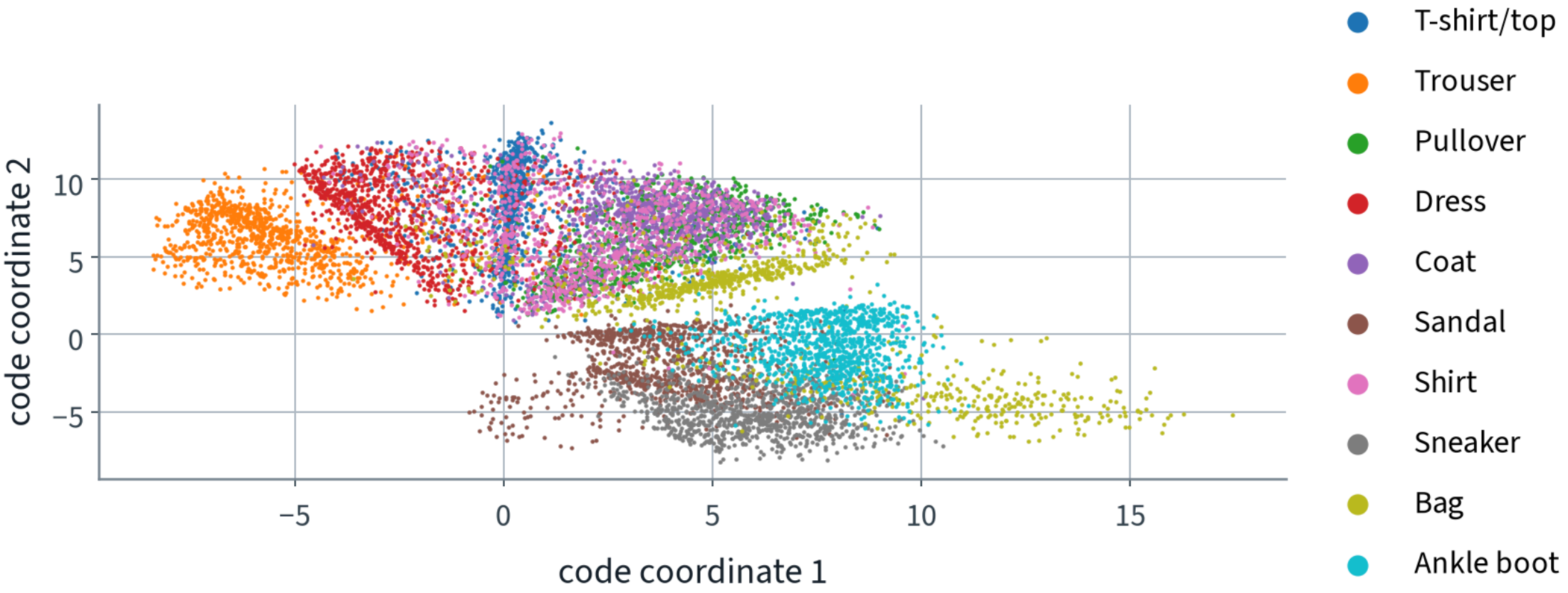
What the loss cannot see

Squared error on pixels is a distance between two arrays. It has no idea which pixels matter.

- **It rewards blur.** The average of a class is close, in pixel distance, to every member of the class. A model short of capacity reconstructs the average and is scored well for it.
- **It is blind to a shift.** Move a sharp edge one pixel and the error is large; smear it over three pixels and the error is smaller. The sharper reconstruction is the worse one, by this loss.
- **It measures pixels, not garments.** A missing button costs nothing; a slightly different grey over the whole sleeve costs a lot.

> **FAILURE CONDITION — A LOSS THAT REWARDS THE AVERAGE**
>
> Reconstruction error falling is not the same as reconstructions improving. Look at them. A code chosen by the loss alone is a code tuned to reproduce the common, and the common is what you already knew.

The code as a picture: $d = 2$



Learned without labels; the colour is the label, added afterwards.

The code as a representation

Thirty-two numbers per garment, computed from the pixels, learned from the pixels. The product questions become geometry in that space:

Question	Operation on the codes	Scored by
similar items	nearest neighbours by Euclidean distance	precision at ten, against the labels
groups	k-means with $k = 10$	mean purity of the clusters, against the labels
the odd ones	reconstruction error as a score	AUROC, against a held-out class

✓ Every one of them has a baseline that needs no learning: the same operation on the 784 raw pixels, and on the 32 PCA scores. A learned code that cannot beat its own input has learned nothing the question needed.

Similar items: nearest neighbours, and how they are scored

```
def precision_at_10(Z):                # Z: one row per catalogue image
    Zq = Z[q]                            # 1,000 fixed queries
    D2 = (Zq**2).sum(1)[:, None] - 2 * Zq @ Z.T + (Z**2).sum(1)[None, :]
    D2[np.arange(len(q)), q] = np.inf          # not your own neighbour
    nn10 = np.argsort(D2, axis=1)[:, :10]
    return (y_test[nn10] == y_test[q][:, None]).mean()
```

- Catalogue: the 10,000 test images; queries: 1,000 of them, each searching the other 9,999
- $\|a-b\|^2 = \|a\|^2 - 2a\cdot b + \|b\|^2$: one matrix product instead of a loop over queries
- **Precision at ten**: of the ten returned, the fraction of the query's class. The label is read here and nowhere else.

One query, ten neighbours, two spaces



Green frames share the query's class: nine of ten in both spaces, and largely the same ten images. The code is a compressed account of pixel distance, not a different opinion about what is similar — which is the finding of the next slide, in one example.

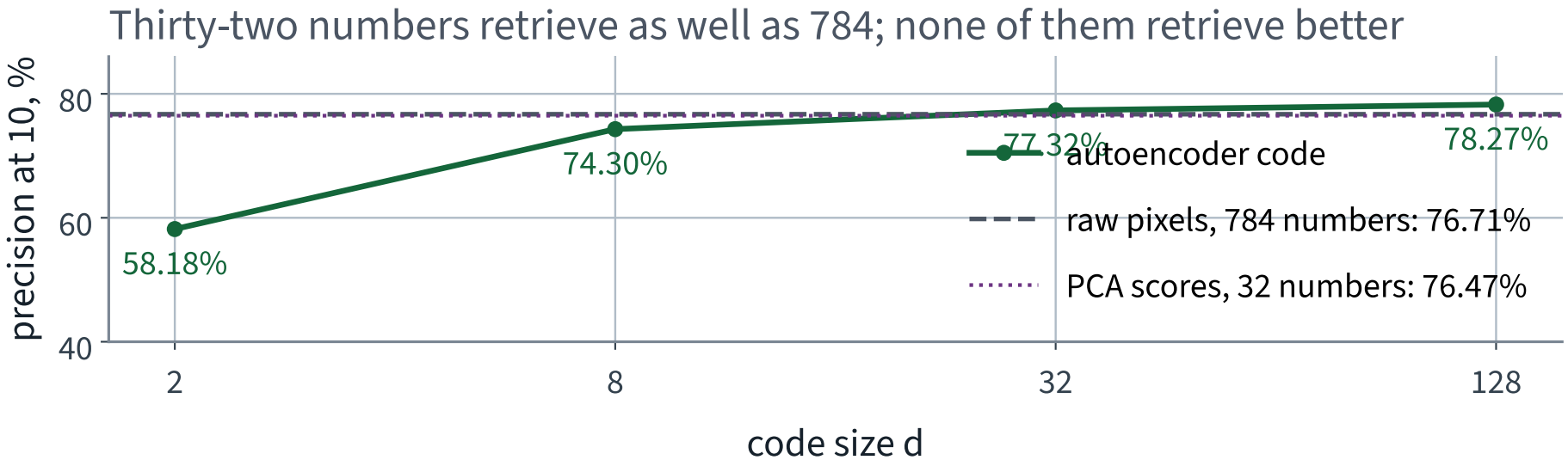
Retrieval, measured

Space	Numbers per item	Precision at 10
Raw pixels	784	76.71%
PCA scores	32	76.47%
Linear autoencoder code	32	77.07%
Autoencoder code	**32**	**77.32% ✓**

Four spaces within a point of one another.

The code did not beat the pixels, and it could not have: nothing in the reconstruction loss is about which garments are *similar*. What it bought is the second column.

How many numbers a garment needs



Two numbers lose a fifth of the precision; eight recover most of it; thirty-two match the pixels; 128 add a point. The representation question is a cost question from eight upwards.

What the code buys: the bill for a catalogue

Per item	Raw pixels	Code	Factor
Numbers stored	784	32	**24.5 ✓**
Multiplications per comparison	784	32	**24.5 ✓**
Bytes, as float32	3,136	128	**24.5 ✓**

A nearest-neighbour search is a comparison against every item in the catalogue. At equal precision, the factor in the last column is the factor on the index's memory and on the query's cost — and for a photograph of real size, 28 × 28 becomes thousands of pixels and the factor grows with it.

The case for a learned representation in a product is, first of all, this table. Quality is the next lecture's problem — and Lecture 23's.

Groups: k-means on the code

Space, $k = 10$	Numbers per item	Mean purity
Raw pixels	784	0.6473
PCA scores	32	0.6471
Autoencoder code	32	0.5891

Purity, from Lecture 6: for each cluster, the share of its members that belong to its commonest class, averaged over the ten clusters. Pixels and PCA scores tie; the learned code loses six points — nothing in the reconstruction loss is about which garments belong together, and here it shows.

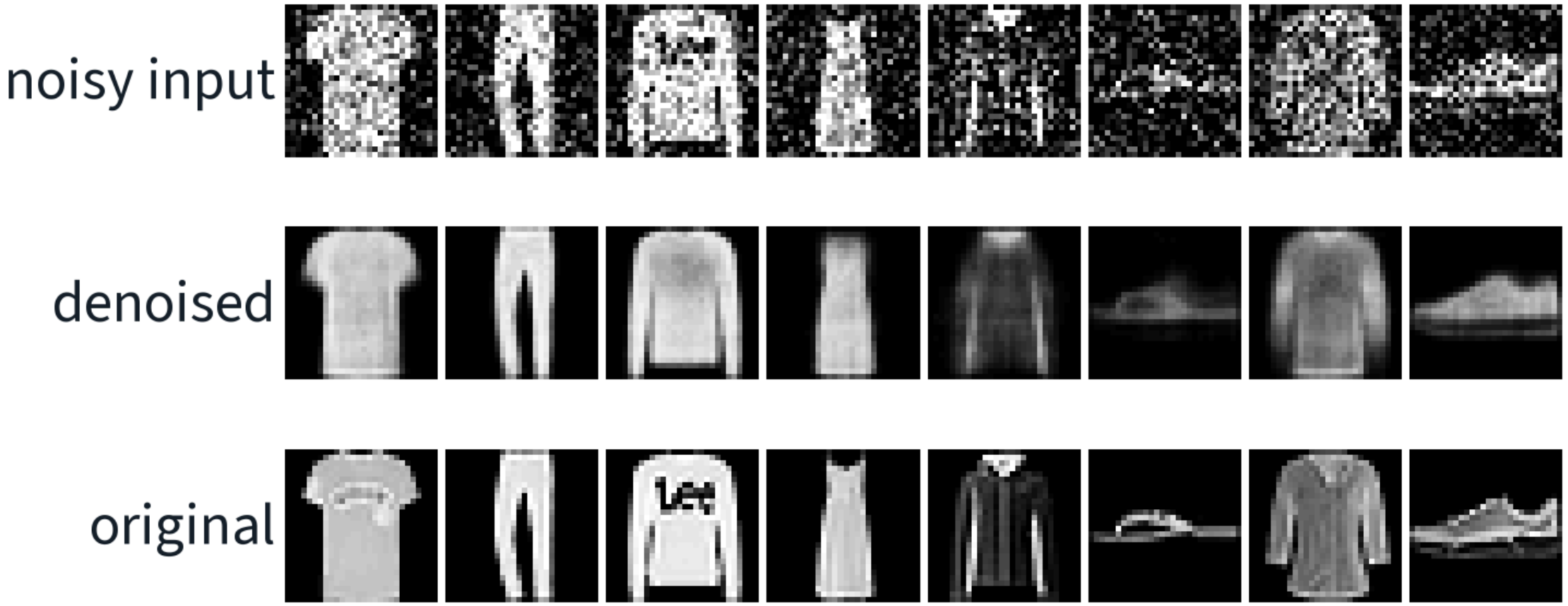
✓ Same lesson as retrieval. The structure k-means finds is the structure the two-dimensional picture drew — footwear, trousers, bags, torso — and it is in the pixels already. The code carries it in 32 numbers, at a price the audit shows.

The denoising autoencoder

```
for xb in batches(X_fit):
    xin  = (xb + 0.3 * torch.randn_like(xb)).clamp(0, 1)   # corrupt the input
    opt.zero_grad()
    loss = F.mse_loss(net(xin), xb)                        # the target is clean
    loss.backward(); opt.step()
```

- The second constraint from the table: what comes in is not what must go out, so copying is impossible even with a wide code
- Fresh noise on every batch — the same corruption twenty times over is a fixed dataset, and can be memorised
- This is Lecture 10's augmentation, with the clean image as the target: a regulariser applied to the *training input only*

Noisy in, garment out



The network never saw a clean input during training. It learned what a garment looks like well enough to put one back — a smooth one, as the squared loss prefers.

Denoising, measured

Error against the clean test image, per pixel	Error
The noisy image itself	✗ 0.05264
The plain autoencoder, fed the noisy image	0.02408
The denoising autoencoder, fed the noisy image	**0.01551 ✓**
The denoising autoencoder, fed the clean image	0.01781

Three things in four rows. The bottleneck alone removes half the noise — noise does not fit through 32 numbers either. Training on corrupted input removes most of the rest. And the price: on clean input the denoiser is a worse autoencoder (0.01781 against 0.01275), and its code retrieves slightly worse, 75.87% against 77.32%.

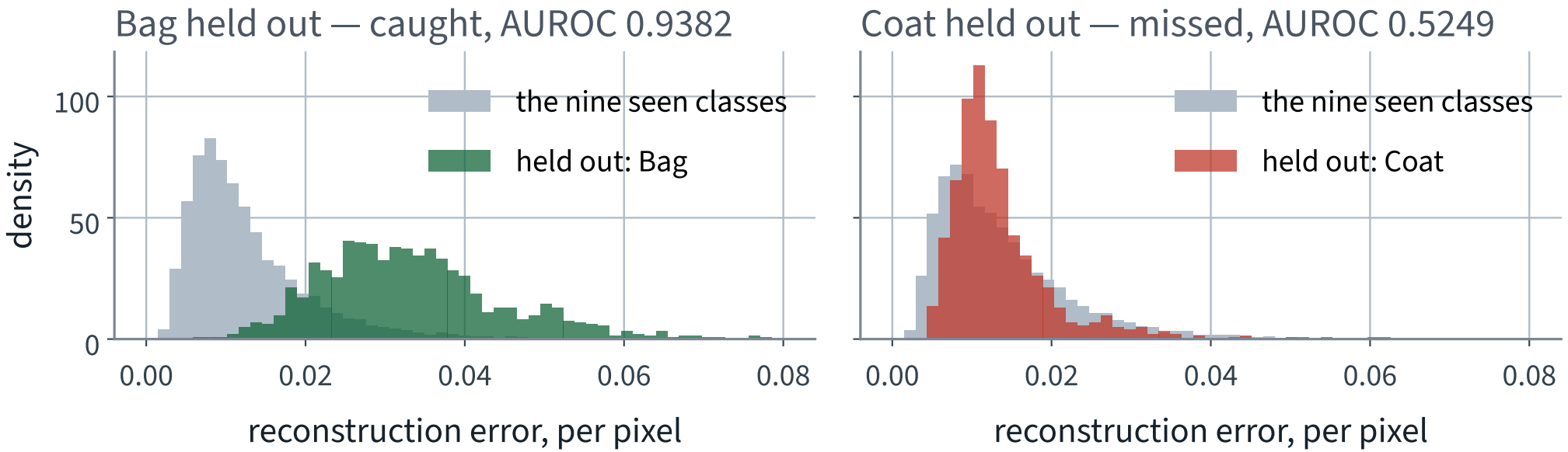
A regulariser is a trade. The constraint that stops copying also stops some fitting.

The odd ones: reconstruction error as a score

- Train on what the catalogue holds. Score every new image by how badly it is reconstructed. An image unlike anything in training should reconstruct badly.
- No labels in training, no examples of the anomaly needed — which is the whole appeal, and the whole risk
- To *measure* it, hold one class out of the fit set, train on the other nine, and ask whether the held-out class scores high

> **AUROC**
>
> The probability that a random held-out image scores above a random seen one. 0.5 is chance; 1.0 is a perfect separation. It needs no threshold, so it measures the score rather than one operating point of it.

Caught, and missed



Bags held out are caught; coats held out are not. A network that learned pullovers, shirts and T-shirts reconstructs a coat as well as it reconstructs them.

Ten held-out classes, ten detectors

Held out	AUROC	Held out	AUROC
Bag ✓	**0.9382 ✓**	Sneaker	0.6081
Trouser	0.8674	Pullover	0.5832
Sandal	0.8664	T-shirt/top	0.5549
Ankle boot	0.7982	Shirt	0.5267
Dress	0.6543	**✗ Coat**	**✗ 0.5249**

The coats' mean error, 0.01345, is *below* the seen classes' 0.01391. The detector would flag a random pullover before a coat.

FAILURE CONDITION — THE SCORE MEASURES IMITABILITY, NOT RARITY

Reconstruction error says how well the training classes can *imitate* an image. A novelty they can imitate is never flagged; an ordinary image they cannot is flagged every time. Whether this detector works is a fact about what the anomalies will look like, and it has to be measured on examples of them.

The three failure conditions, in one place

Condition	What happens	Where it was measured
No constraint on the code	the identity is reachable; zero loss, nothing learned	the floor at $d = 784$ is 0
A trained approximation against an exact optimum	at a wide bottleneck the network sits above the SVD	$d = 128$: 0.01179 against 0.00641
Reconstruction error as an anomaly score	an imitable novelty is never flagged	coats: AUROC 0.5249

And one that is a property of the loss rather than of the method: squared error rewards the average, so a falling loss is not an improving reconstruction until you have looked.

The variational autoencoder

The autoencoders so far give one code per image and nothing in between: decode a point between two codes and there is no reason the output is a garment. The **variational autoencoder** makes the encoder output a *distribution*, $\mathcal{N}(\boldsymbol{\mu}(\mathbf{x}), \operatorname{diag}\ \boldsymbol{\sigma}^2(\mathbf{x}))$, samples a code from it, and adds a term that keeps every such distribution close to $\mathcal{N}(\mathbf{0}, \mathbf{I})$:

BEYOND THE SYLLABUS, FOR CONTEXT

$$L = \underbrace{\|\mathbf{x} - \hat{\mathbf{x}}\|^2}_{\text{reconstruction}} + \underbrace{\frac{1}{2}\sum_{j=1}^{d}\left(\sigma_j^2 + \mu_j^2 - 1 - \log \sigma_j^2\right)}_{\text{KL to the standard normal}}$$

The second term is a regulariser on the code — the third row of the table on slide 8 — and it fills the code space: a code drawn from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ decodes to something plausible, which makes the decoder a *generator*.

The reparametrisation trick

A sampled code breaks the chain rule of Lecture 8: there is no derivative of a random draw with respect to the parameters that chose its distribution. The repair moves the randomness outside the graph: BEYOND THE SYLLABUS, FOR CONTEXT

$$\mathbf{z} = \boldsymbol{\mu}(\mathbf{x}) + \boldsymbol{\sigma}(\mathbf{x}) \odot \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

- $\boldsymbol{\varepsilon}$ is drawn first and treated as an input; $\mathbf{z}$ is then a deterministic function of $\boldsymbol{\mu}$, $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$
- Reverse mode differentiates through $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ as through any layer; the draw is never differentiated

Not examined. It is here because every generative model you will meet from Lecture 24 onwards contains this move, and because it is one line.

Where this leads: representations, and what trains them

Representation	Trained by	What distances in it mean	Lecture
PCA scores	the SVD	pixel distance, in the best d directions	6
autoencoder code	**reconstructing the input**	**pixel distance, through a curved map, in a basis nobody chose**	**11**
a classifier's penultimate layer	labels	whatever separates the classes	13
a dense-retrieval embedding	query–document pairs	relevance	20
a contrastive embedding	pairs that belong together	similarity, as the pairs defined it	23

✓ Today's code matched the pixels on retrieval and could not beat them, because the loss never saw a pair. The rest of the course is, in part, the story of what happens when it does.

The notebook for this lecture

Open Lecture 11 in Colab

- **Runs on free CPU**, and on the CPU by choice: 12,000 images and small networks, and the numbers on these slides are its run
- The floor from the SVD, then the linear autoencoder trained against it: loss, test error, principal angles, the R^2 of the code on the PCA scores
- The non-linear autoencoder, the sweep over d, the reconstructions, the two-dimensional code drawn
- Precision at ten and k-means in four spaces; the denoiser; ten anomaly detectors, one per held-out class

> **WHAT TO CHANGE**
>
> Set `NOISE = 0` in the denoising section and run again. The denoising rows become the plain autoencoder's, to the digit — they were always the same model with one line changed.

Design exercise · in the manner of an ML system-design interview

The task: “see similar” on a fashion marketplace

> **THE BRIEF**
>
> An online fashion marketplace sells garments listed by thousands of independent brands and shops. The product is its shopping app; the feature is a **“see similar”** strip under every item page, and a camera search that finds catalogue items like a photo the shopper took. The users are shoppers; the customers are the sellers, who pay a commission on every sale. The business measures the strip’s click-through, the conversion it adds, and the return rate of what it sells.

- **Given:** millions of seller photographs with an unreliable category field, the shoppers’ click and purchase logs, the sellers’ stock and price feeds
- **Consumers of the output:** the strip, the camera search, and the screening that keeps non-garments out of the catalogue
- **Regulators:** data protection, for the shoppers’ photographs; consumer protection and brand owners, for what “similar” may show

✓ Nine steps follow, each with its pitfall; the model at the centre is today’s code, learned from the photographs

Step 1 — Frame it honestly: a ranking, scored as one

- The output is a ranked list of catalogue items for one query image; the metric is precision at the strip's length on held-out items, read beside its anchor: the same strip filled from the query's category at random
- The label for scoring comes from what shoppers did — items viewed or bought in the same session — cleaned of the strip's own clicks; the seller's category field is a weak second source
- The baseline before any model: the same category, in stock, most popular first

> **PITFALL**
>
> Reporting the reconstruction error. It is the training signal, and the shopper never sees it: a code can reconstruct every photograph beautifully and return the wrong garments. Score the ranking the product shows, and report it against the anchor, or the first number will say nothing about the feature.

Step 2 — The photographs are not as alike as the dataset's

Pitfall	What happens	What to do
Studio, not garment	sellers shoot on different backgrounds, at different sizes, on or off a model; a code learned from the pixels learns the studio, and the strip returns the same seller's other items	crop to the garment, resize, normalise the background before anything is fitted; check the strip for seller clumping
Shoppers' photographs	a camera-search image is personal data, and is nothing like a studio shot	encode on the device or discard after encoding; never keep the photo; train the encoder with camera-style augmentations
Duplicates	the same garment listed by four sellers, and in six colours	link variants to one item before splitting, or the split leaks

✓ Write down what one "item" is before the first fit. Every later number is a number about *that* unit.

Step 3 — Split by item, and by seller

- Hold out whole items — every photograph and variant of them — and hold out whole sellers for a second check: the feature has to work for a shop that joined this morning
- Choose the code size, the distance and the ranking rules on the validation items; read the test items once

> **PITFALL**
>
> A random split over photographs puts the red and the blue of one dress on both sides. The nearest neighbour of a held-out photo is then its own item in another colour, precision looks superb, and the number says nothing about a garment the catalogue has never seen. Split on the item identifier from step 2, and keep a seller-level split beside it.

Step 4 — Learn the code from the catalogue, and size it by the ranking

- Train the autoencoder on the whole catalogue of standardised photographs — no labels needed, so every listing counts, and a convolutional encoder once the photographs are real-sized
- Read the code from the encoder alone; sweep its size and pick the smallest that holds the validation precision — the cost table says why smallest
- Denoising and augmentation as the regulariser: the same garment, shifted, in another colour, on another background, must land on the same code

> **PITFALL**
>
> Choosing the code size by reconstruction error. It falls at every increase, forever; the ranking metric flattens early and the index cost does not. A code sized by the loss is a code sized by the wrong question, and the bill for it is paid on every query.

Step 5 — The index: codes stored once, searched in milliseconds

- Encode every catalogue item once, when it is listed, and store the code beside it; the query is encoded at request time and compared against the stored codes, never against the photographs
- Exact search is a matrix product at a hundred thousand items; at millions, an approximate-neighbour index trades a little recall for a hundred-fold speed, and that recall is measured, not assumed
- Normalise the codes to unit length and compare by cosine, so that a bright photograph is not far from a dark one of the same dress

> **PITFALL**
>
> A distance nobody chose. The code’s coordinates are in a basis the optimiser landed on, and one coordinate with a large range dominates the Euclidean distance. Normalise, or whiten against the validation set, and check that the strip’s neighbours do not all share one bright background.

Step 6 — Similarity is one feature of the ranking, not the ranking

- The strip ranks the nearest hundred by a second model that sees similarity, stock, size availability, price band, the seller’s return rate and the shopper’s history — a table, and a boosted ensemble on it
- Business rules sit on top: in stock only, the shopper’s size, nothing from a seller under review

> **PITFALL — THE LOOK-ALIKE UNDER A BRAND’S PAGE**
>
> Visual similarity is blind to who made the garment. Under a luxury brand’s listing the strip shows the imitations that look most like it, the brand — a paying customer — sees its page selling look-alikes, and the counterfeit complaints land with the regulator. Brand and seller constraints belong in the ranking layer, and the similarity model must never be the last word.

Step 7 — Screening uploads by reconstruction error

- The same encoder–decoder scores every new upload: a screenshot, a blank, a photograph of a receipt reconstructs badly and goes to a review queue before it is listed
- Calibrate the threshold per category on the validation items, and measure the queue's precision with the reviewers' decisions

> **PITFALL**
>
> The score follows what the catalogue can imitate. A plain photograph of nothing reconstructs well and passes; an elaborate, legitimate gown reconstructs badly and is flagged, and its seller waits a day for a listing. Route to review, never reject automatically, and read the detector's AUROC on the kinds of upload it is meant to catch.

Step 8 — Release, monitor, retrain

- Release behind an A/B test on live traffic, read on the strip's click-through, the conversion it adds and the return rate of what it sells, with the sellers' exposure held fair across both arms
- Monitor the distribution of codes as the catalogue turns over with the seasons; retrain on a schedule and re-encode the whole catalogue in one atomic swap of the index
- Keep a randomised slot in the strip, so that the clicks the next model trains on are not only the ones this model chose

> **PITFALL**
>
> Two encoders in one index. A retrained encoder puts the same garment at a different code, so an index holding codes from both versions compares numbers that no longer mean the same thing, and the strip fills with garments that are near nothing. Version the encoder with the index, and never serve a query from one against codes from the other.

Step 9 — Ship it: pipeline and pitfalls

Standardised photographs, variants linked into items → split by item and seller → autoencoder, code sized on validation precision → normalised codes, versioned index → ranking with business rules → review queue → A/B test → retrain, re-encode, monitor.

Pitfall	Address it by
Reconstruction error as the product metric	precision on the strip against the category anchor
A code that learns the studio	standardised photographs; seller clumping checked
Variants across the split	splitting by item and by seller
Code sized by the loss	the smallest size that holds validation precision
A distance nobody chose	normalised codes, cosine, whitening
Look-alikes under a brand	brand and seller rules in the ranking layer
The screen flags the elaborate, passes the blank	a review queue, calibrated per category, measured
Two encoders in one index	versioned encoder and index, atomic re-encoding

What we did

1. Trained a network with no labels: compress to a code, reconstruct, and make the bottleneck the constraint — a regulariser in Lecture 10's sense, because without it the identity is reachable
2. Derived that a linear autoencoder with squared loss cannot beat the projection onto the principal subspace, and reaches it: the floor is $\sum_{j>d} \sigma_j^2$, the subspace is Lecture 6's, the code is unique only up to a change of basis — and checked all three numerically
3. Built the non-linear one: a sixth below the floor at $d = 32$, and above the exact projection at $d = 128$, because a trained approximation is not an exact optimum
4. Read the code as a representation: 32 numbers retrieve and cluster as well as 784 pixels, and no better, at a twenty-fourfold saving — the loss never saw a pair
5. Denoising as a constraint on the input; reconstruction error as an anomaly score that catches bags and misses coats, because it measures imitability
6. Designed a marketplace's "see similar" feature in nine steps — a ranking metric against its anchor, standardised photographs, a split by item, a code sized by precision, a normalised versioned index, business rules above similarity, a review queue, an A/B test — each named for the pitfall it avoids

NOT PRESENTED — FOR AFTERWARDS

Exercises Lecture 11

five, in the style of the written paper

Exercises — Lecture 11

1 A linear autoencoder $784 \to d \to 784$ with squared loss is trained to convergence on centred data. State the smallest error it can reach, in terms of the singular values of the centred data, and name the subspace through which it reconstructs. [4]

2 Two linear autoencoders are trained from different seeds to the same loss, and their codes for the same image differ. State whether this is a bug, and why. [4]

3 The code size is set equal to the number of pixels and nothing else constrains the network. State what the trained network computes, and what its code is worth as a representation. [3]

Solutions on Lecture 12's deck.

Exercises — Lecture 11 (continued)

4 An anomaly detector scores images by their reconstruction error from an autoencoder trained on nine classes; on the tenth, held out, it scores at chance. Explain the mechanism, and say what would be needed to know in advance whether such a detector works. [5]

5 Nearest-neighbour retrieval in a 32-number autoencoder code scores the same precision at ten as in the 784 raw pixels. State what the code bought, and why it could not have bought more. [4]

Solutions on Lecture 12’s deck.

THE FIVE SET AT THE END OF LECTURE 10

Solutions Lecture 10

not part of this lecture

Solutions — Lecture 10

1. A hidden unit's activations a_j feed a pre-activation $z = \sum_j w_j a_j$. Dropout keeps each unit with…

✓ $\mathbb{E}[\tilde{z}] = \sum_j \mathbb{E}[m_j]\, w_j a_j = (1-p)\, z$; inverted dropout divides the kept activations by $1-p$ during training, so $\mathbb{E}[\tilde{z}] = z$ and the unmasked network at test time is the expectation.

- The masks are independent Bernoulli$(1-p)$ and the weights are fixed, so the expectation passes through the sum
- Without the division every pre-activation at test time is $1/(1-p)$ times larger than anything the weights were trained on

2. Dropout preserves the expected pre-activation. Show that it does not preserve the second moment, give the…

✓ $\mathbb{E}[(m_j a_j/(1-p))^2] = a_j^2/(1-p)$ because $m_j^2 = m_j$: each dropout layer multiplies the second moment by $1/(1-p)$, so twenty layers multiply it by $(1-p)^{-20}$ — dropout goes on the top layer or two, not on every layer.

- $m_j \in \{0, 1\}$, so $\mathbb{E}[m_j^2] = \mathbb{E}[m_j] = 1-p$, and dividing by $(1-p)^2$ leaves $1/(1-p)$
- A factor applied once is a regulariser; the same factor applied twenty times is noise the network must learn through

Solutions — Lecture 10 (2)

3. Add $\frac{\lambda}{2}\|\theta\|^2$ to the loss and write one gradient-descent step. Then say what…

✓ $\theta \leftarrow \theta - \eta(\nabla L + \lambda\theta) = (1 - \eta\lambda)\,\theta - \eta\nabla L$: shrink every weight by $1 - \eta\lambda$, then step. Inside Adam the $\lambda\theta$ term is divided by the running gradient scale like everything else, so the decay is scaled away; AdamW applies the factor $1 - \eta\lambda$ outside the adaptive step.

- The gradient of $\frac{\lambda}{2}\|\theta\|^2$ is $\lambda\theta$
- Adam divides each coordinate's step by its gradient history, so a

4. A colleague trains for fifty epochs, keeps the weights of the epoch with the highest validation accuracy, and…

✓ Right: the epoch is chosen on the validation set, and the best epoch's weights are kept rather than the last. Wrong: the reported number is the one that chose the epoch, so it is optimistic; report the test accuracy of the kept weights, read once.

- The maximum over fifty validation readings is biased upward by the selection
- The test set exists to be read after every choice has been made

Solutions — Lecture 10 (3)

5. Six values of a weight-decay knob are tried, each run read at its best validation epoch. The value with the…

✓ $t_t - t_v \geq 0$ always, with equality when the two choices coincide; the bias is the gap between the maximum of several test readings and one test reading, and it is invisible from a single number because both are accuracies on the same set.

- t_t is by construction the largest test accuracy in the table, so it cannot be below any other entry, t_v included
- More candidates can only raise a maximum, so the bias grows with the size of the search and nothing in the reported figure records that size

LECTURE 12 · GÉRON CH. 12

Convolutional networks

Why a dense layer is the wrong shape for an image
Weight sharing and equivariance, derived
A network that sees Flowers102 at its real size

Run the Lecture 11 notebook first